

Zero-Copy ASharedMemory & UNIX Domain Socket IPC

1. Bypassing Android IPC Bottlenecks

Standard Android IPC (Binder) enforces a strict 1MB transaction limit and requires repeated CPU copying.

- **ASharedMemory Allocation**: Anonymous shared memory regions allocated via `ASharedMemory_create`` and mapped into process spaces via `mmap``.
- **Abstract UNIX Domain Sockets**: Sockets initialized with `sun_path[0] = "\0`` bind directly to Linux's abstract network namespace, eliminating physical disk I/O and filesystem permission blocks.
- **File Descriptor Handshake**: File descriptors are passed seamlessly between daemons using POSIX ancillary messages (`SCM_RIGHTS``).